



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 01

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5 , CSS3 , JavaScript , Visual Studio Code , Google Chrome

Experiment Program Code:

1. External CSS:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Experiment 1</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
  <div>
```

```
    <h1>Welcome to the Lab!</h1>
```

```
    <p><strong>Name:</strong>Tejas Bawane</p>
```

```
    <p><strong>PRN:</strong> 24070521147</p>
```

```
    <button onclick="alert('External CSS active!')">Test External CSS</button>
```

```
  </div>
```



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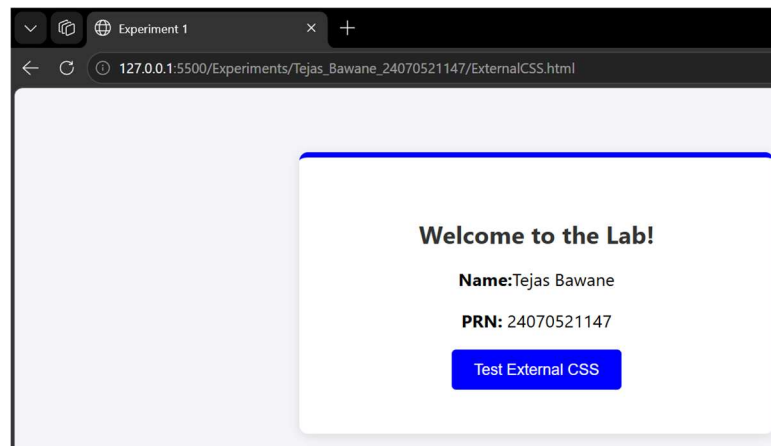
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</body>

</html>

Output:



2. Inline CSS:

<!DOCTYPE html>

<html>

<head>

<title>Experiment 1</title>

</head>

<body style="font-family: 'Segoe UI', sans-serif; background-color: #f4f4f9; display: flex; justify-content: center; padding-top: 50px; margin: 0;">

<div style="background-color: white; padding: 40px; border-radius: 8px; border-top: 5px solid blue; box-shadow: 0 2px 10px rgba(0,0,0,0.1); text-align: center; width: 350px;">

<h1 style="color: #333; font-size: 22px;">Welcome to the Lab!</h1>



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<p>Name: Tejas Bawane</p>

<p>PRN: 24070521147</p>

<button onclick="alert('Inline CSS active!')"

style="padding: 10px 20px; background-color: blue; color: white; border: none;
border-radius: 4px; cursor: pointer; font-size: 14px;">

Test Inline CSS

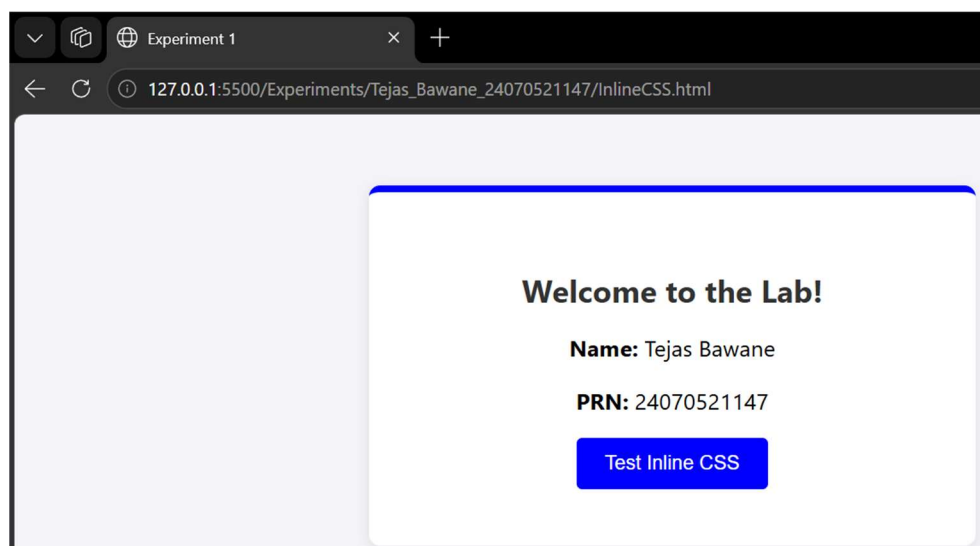
</button>

</div>

</body>

</html>

Output:





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3. Internal CSS:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Experiment 1</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: 'Segoe UI', sans-serif;
```

```
      background-color: #f4f4f9;
```

```
      display: flex;
```

```
      justify-content: center;
```

```
      padding-top: 50px;
```

```
      margin: 0;
```

```
    }
```

```
    div {
```

```
      background-color: white;
```

```
      padding: 40px;
```

```
      border-radius: 8px;
```

```
      border-top: 5px solid blue;
```

```
      box-shadow: 0 2px 10px rgba(0,0,0,0.1);
```



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```
text-align: center;

width: 350px;

}

h1 {

  color: #333;

  font-size: 22px;

}

button {

  padding: 10px 20px;

  background-color: blue;

  color: white;

  border: none;

  border-radius: 4px;

  cursor: pointer;

  font-size: 14px;

}

button:hover {

  background-color: darkblue;

}
```



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</style>

</head>

<body>

<div>

<h1>Welcome to the Lab!</h1>

<p>Name: Tejas Bawane</p>

<p>PRN: 24070521147</p>

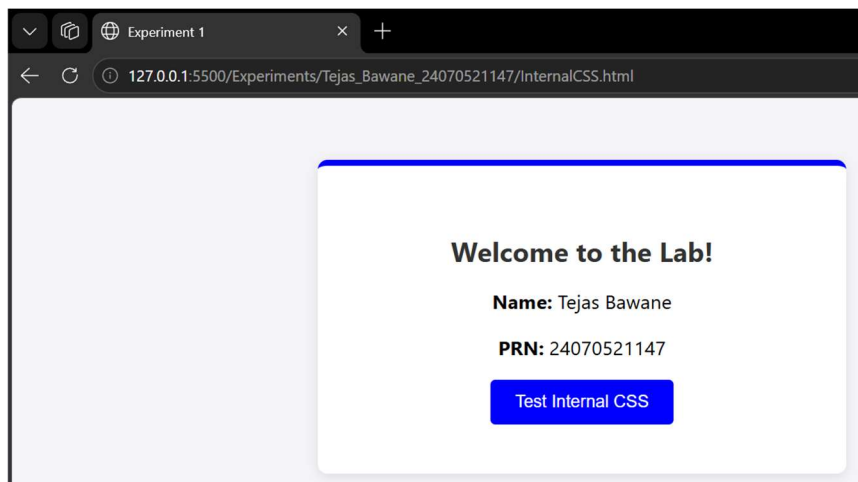
<button onclick="alert('Internal CSS active!')">Test Internal CSS</button>

</div>

</body>

</html>

Output:





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Style:

/* External CSS used to style the page */

body {

font-family: 'Segoe UI', sans-serif;

background-color: #f4f4f9;

display: flex;

justify-content: center;

padding-top: 50px;

}

div {

background-color: white;

padding: 40px;

border-radius: 8px;

border-top: 5px solid blue; /* Adds a nice accent */

box-shadow: 0 2px 10px rgba(0,0,0,0.1);

text-align: center;

width: 350px;

}

h1 {



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color: #333;

font-size: 22px;

}

button {

padding: 10px 20px;

background-color: blue;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 14px;

}

button:hover {

background-color: darkblue;

}



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Case Study No.: 01

Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

Software/Tools Required: HTML5 , CSS3 , JavaScript , Visual Studio Code , Google Chrome

Experiment Program Code:

1. Main:

```
<html>
```

```
<head>
```

```
<title>SIT Nagpur - Department Page</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    margin: 40px;
```

```
    background-color: #f4f4f9;
```

```
    color: #333;
```

```
}
```

```
.container {
```

```
    max-width: 600px;
```

```
    background: #fff;
```

```
    padding: 25px;
```



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```
border-radius: 8px;

box-shadow: 0 2px 5px rgba(0,0,0,0.1);

}

h1 {

    color: #1a365d;

}

.btn {

    display: inline-block;

    margin-top: 15px;

    padding: 10px 15px;

    background-color: #3182ce;

    color: #fff;

    text-decoration: none;

    border-radius: 5px;

}

.btn:hover {

    background-color: #2b6cb0;

}

</style>

</head>

<body>
```



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```
<div class="container">
```

```
<h1>Department of Computer Science & Engineering</h1>
```

```
<p>Welcome to the official department portal of SIT Nagpur. We offer state-of-the-art facilities and top-tier academic guidance to shape the innovators of tomorrow.</p>
```

```
<!-- Linking the student information page -->
```

```
<a href="CS1_Stud.html" class="btn">View Student Information</a>
```

```
</div>
```

```
<!-- JavaScript Program to display the Alert Box -->
```

```
<script>
```

```
    window.onload = function() {
```

```
        alert("Welcome to SIT Nagpur");
```

```
    };
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



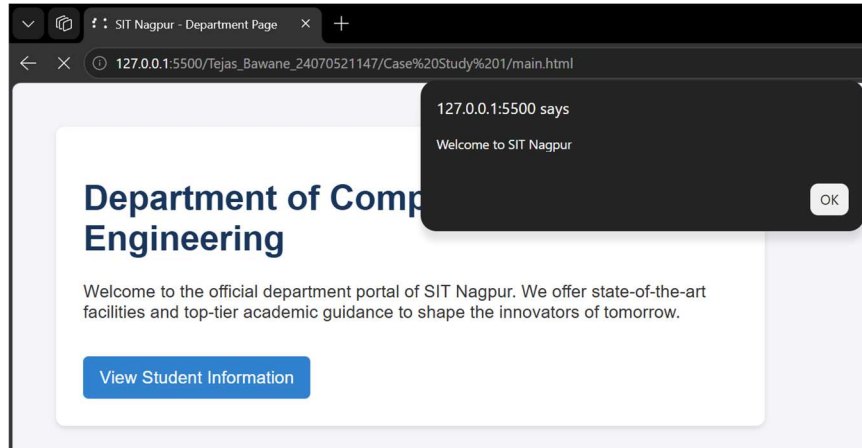
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2. Student:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Student Information</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
margin: 40px;
```

```
background-color: #f4f4f9;
```

```
color: #333;
```

```
}
```

```
.container {
```

```
max-width: 600px;
```



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```
background: #fff;

padding: 25px;

border-radius: 8px;

box-shadow: 0 2px 5px rgba(0,0,0,0.1);

}

h2 {

    color: #1a365d;

}

table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;

}

th, td {

    padding: 10px;

    border: 1px solid #ddd;

    text-align: left;

}

th {

    background-color: #edf2f7;

}

.back-link {
```



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```
display: inline-block;

margin-top: 20px;

color: #3182ce;

text-decoration: none;

}

.back-link:hover {

    text-decoration: underline;

}

</style>

</head>

<body>

<div class="container">

    <h2>Student Information Page</h2>

    <table>

        <thead>

            <tr>

                <th>Roll No</th>

                <th>Student Name</th>

                <th>Year</th>

            </tr>
```



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</thead>

<tbody>

<tr>

<td>24070521147</td>

<td>Tejas Bawane</td>

<td>3rd Year</td>

</tr>

</tbody>

</table>

<!-- Link back to Department page -->

← Back to Department Page

</div>

</body>

</html>

Output:



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SIT Nagpur - Department Page

Student Information

127.0.0.1:5500/Tejas_Bawane_24070521147/Case%20Study%201/stud.html

Student Information Page

Roll No	Student Name	Year
24070521147	Tejas Bawane	3rd Year

[← Back to Department Page](#)



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Experiment No.: 02

Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Billing Calculator</title>
```

```
</head>
```

```
<body>
```

```
<h2>Billing Calculator</h2>
```

```
<script>
```

```
const GST = 0.18;
```



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```
var customerName = prompt("Enter Customer Name: ");
```

```
var productName = prompt("Enter Product Name: ");
```

```
var quantity = Number(prompt("Enter quantity"));
```

```
let price = Number(prompt("Enter price per Item"));
```

```
let total = quantity * price;
```

```
let gstAmount = total * GST;
```

```
let finalBill = total + gstAmount;
```

```
const bill = {
```

```
    customer: customerName,
```

```
    product: productName,
```

```
    quantity,
```

```
    price,
```

```
    total,
```

```
    gstAmount,
```

```
    finalBill
```

```
};
```



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```
const {  
  
    customer,  
  
    product,  
  
    gstAmount: gst,  
  
    finalBill: amount  
}  
= bill;  
  
document.write(`<h3>Billing Receipt</h3>`);  
  
document.write(`<p>Customer: ${customer}</p>`);  
  
document.write(`<p>Product: ${product}</p>`);  
  
document.write(`<p>Quantity: ${quantity}</p>`);  
  
document.write(`<p>Price per Item: ₹${price}</p>`);  
  
document.write(`<p>Total: ₹${total}</p>`);  
  
document.write(`<p>GST (18%): ₹${gst.toFixed(2)}</p>`);  
  
document.write(`<h3>Final Bill: ₹${amount.toFixed(2)}</h3>`);  
  
</script>
```



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</body>

</html>

Output:

Billing Calculator

Billing Receipt

Customer: Tejas

Product: Laptop

Quantity: 2

Price per Item: ₹80000

Total: ₹160000

GST (18%): ₹28800.00

Final Bill: ₹188800.00



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Case Study No.: 02

Title: To design and develop an interactive, client-side billing application (BILL.HTML) using HTML5, CSS3, and JavaScript (ES6+) that parses real-time product quantities, applies discount thresholds, calculates taxation rates, and renders an itemised invoice directly onto the Document Object Model (DOM).

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>E-Commerce Automated Billing Engine</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
      line-height: 1.6;
```

```
      margin: 0;
```

```
      padding: 30px;
```



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```
background-color: #f7f9fc;

}

.billing-container {

    max-width: 700px;

    background: #ffffff;

    margin: auto;

    padding: 25px;

    border-radius: 8px;

    box-shadow: 0 4px 15px rgba(0,0,0,0.05);

}

h2, h3 {

    color: #1a365d;

    text-align: center;

    margin-top: 0;

}

.student-badge {

    text-align: center;

    font-size: 14px;

    color: #718096;
```



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margin-bottom: 25px;

border-bottom: 2px dashed #e2e8f0;

padding-bottom: 10px;

}

.form-row {

display: flex;

gap: 15px;

margin-bottom: 15px;

}

.form-group {

flex: 1;

display: flex;

flex-direction: column;

}

label {

font-weight: 600;

margin-bottom: 6px;

color: #2d3748;

}



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```
input, select {  
  
    padding: 10px;  
  
    border: 1px solid #cbd5e0;  
  
    border-radius: 5px;  
  
    font-size: 15px;  
  
}  
  
input:focus, select:focus {  
  
    outline: none;  
  
    border-color: #3182ce;  
  
}  
  
button {  
  
    background: #2b6cb0;  
  
    color: #ffffff;  
  
    padding: 12px;  
  
    border: none;  
  
    border-radius: 5px;  
  
    cursor: pointer;  
  
    width: 100%;  
  
    font-size: 16px;
```



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```
font-weight: bold;

margin-top: 10px;

transition: background 0.2s;

}

button:hover {

    background: #2c5282;

}

.invoice-box {

    margin-top: 25px;

    padding: 20px;

    background: #f8fafc;

    border: 1px dashed #4299e1;

    border-radius: 6px;

    display: none;

}

.invoice-table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;
```



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}

.invoice-table th, .invoice-table td {

padding: 10px;

text-align: left;

border-bottom: 1px solid #e2e8f0;

}

.invoice-table th {

background: #edf2f7;

color: #2d3748;

}

.text-right {

text-align: right !important;

}

.total-highlight {

font-size: 18px;

font-weight: bold;

color: #e53e3e;

}

</style>



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</head>

<body>

<div class="billing-container">

<h2>E-Commerce Billing System</h2>

<div class="student-badge">Developer: Tejas Bawane | PRN: 24070521147</div>

<form id="billingForm" onsubmit="generateInvoice(event)">

<div class="form-row">

<div class="form-group">

<label for="customerName">Customer Name:</label>

<input type="text" id="customerName" placeholder="Enter name" required>

</div>

<div class="form-group">

<label for="invoiceNo">Invoice Number:</label>

<input type="text" id="invoiceNo" placeholder="INV-2026-001" required>

</div>

</div>



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```
<div class="form-row">
```

```
<div class="form-group" style="flex: 2;">
```

```
<label for="itemName">Item Description:</label>
```

```
<input type="text" id="itemName" placeholder="Product name/model" required>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="itemCategory">Category:</label>
```

```
<select id="itemCategory">
```

```
<option value="Electronics">Electronics (18% GST)</option>
```

```
<option value="Clothing">Clothing (5% GST)</option>
```

```
<option value="Essentials">Essentials (0% GST)</option>
```

```
</select>
```

```
</div>
```

```
</div>
```

```
<div class="form-row">
```

```
<div class="form-group">
```

```
<label for="unitPrice">Unit Price (₹):</label>
```



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<input type="number" id="unitPrice" step="0.01" min="0" placeholder="0.00"
required>

</div>

<div class="form-group">

<label for="quantity">Quantity:</label>

<input type="number" id="quantity" min="1" placeholder="1" required>

</div>

<div class="form-group">

<label for="discountRate">Discount (%):</label>

<input type="number" id="discountRate" min="0" max="100" placeholder="0">

</div>

</div>

<button type="submit">Compute Calculations & Generate Receipt</button>

</form>

<div id="invoiceBox" class="invoice-box">

<h3>RETAIL INVOICE RECEIPT</h3>

<p>Customer Reference:



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Invoice String ID: </p>

<table class="invoice-table">

<thead>

<tr>

<th>Item Description</th>

<th class="text-right">Rate</th>

<th class="text-right">Qty</th>

<th class="text-right">Subtotal</th>

</tr>

</thead>

<tbody>

<tr>

<td> ()</td>

<td class="text-right">₹</td>

<td class="text-right"></td>

<td class="text-right">₹</td>

</tr>

<tr>



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<td colspan="3" class="text-right">Discount Applied:</td>

<td class="text-right" style="color: #38a169;">-₹</td>

</tr>

<tr>

<td colspan="3" class="text-right">Calculated Tax Component (GST):</td>

<td class="text-right">₹</td>

</tr>

<tr style="background: #f7fafc;">

<td colspan="3" class="text-right total-highlight">Grand Total
Payable:</td>

<td class="text-right total-highlight">₹</td>

</tr>

</tbody>

</table>

</div>

</div>

<script>

function generateInvoice(event) {



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```
event.preventDefault();
```

```
// 1. Data Capture Extractors
```

```
var customer = document.getElementById("customerName").value.trim();
```

```
var invoiceNum = document.getElementById("invoiceNo").value.trim();
```

```
var item = document.getElementById("itemName").value.trim();
```

```
var category = document.getElementById("itemCategory").value;
```

```
var price = parseFloat(document.getElementById("unitPrice").value);
```

```
var qty = parseInt(document.getElementById("quantity").value);
```

```
var discRate = parseFloat(document.getElementById("discountRate").value) || 0;
```

```
// 2. Arithmetic Matrix Calculations
```

```
var rawSubtotal = price * qty;
```

```
var discountAmount = (discRate / 100) * rawSubtotal;
```

```
var discountedSubtotal = rawSubtotal - discountAmount;
```

```
// Tax Percentage Matrix Assignment
```

```
var taxPercentage = 0;
```

```
if (category === "Electronics") taxPercentage = 18;
```



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```
else if (category === "Clothing") taxPercentage = 5;

else if (category === "Essentials") taxPercentage = 0;


var taxAmount = (taxPercentage / 100) * discountedSubtotal;

var grandTotal = discountedSubtotal + taxAmount;


// 3. UI Node Rendering Updates

document.getElementById("invCustomer").innerText = customer;

document.getElementById("invId").innerText = invoiceNum;

document.getElementById("invItem").innerText = item;

document.getElementById("invCat").innerText = category;

document.getElementById("invPrice").innerText = price.toFixed(2);

document.getElementById("invQty").innerText = qty;

document.getElementById("invSub").innerText = rawSubtotal.toFixed(2);

document.getElementById("invDisc").innerText = discountAmount.toFixed(2);

document.getElementById("invTax").innerText = taxAmount.toFixed(2);

document.getElementById("invGrand").innerText = grandTotal.toFixed(2);


// Slide/Toggle element block presence configurations
```



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```
document.getElementById("invoiceBox").style.display = "block";

}

</script>

</body>

</html>
```

Output:

E-Commerce Billing System
Developer: Tejas Bawane | PRN: 24070521147

Customer Name: Invoice Number:

Item Description: Category:

Unit Price (₹): Quantity: Discount (%):

Compute Calculations & Generate Receipt

RETAIL INVOICE RECEIPT

Customer Reference: Tejas
Invoice String ID: INV-2026-001

Item Description	Rate	Qty	Subtotal
Laptop (Electronics)	₹80000.00	1	₹80000.00
Discount Applied:			-₹20000.00
Calculated Tax Component (GST):			₹10800.00
Grand Total Payable:			₹70800.00



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Experiment No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>

<head>

<title>JS Practical - Multi-Subject Grading System</title>

<style>

body {

    font-family: Arial, sans-serif;

    max-width: 480px;

    margin: 30px auto;

    padding: 20px;

    border: 1px solid #ccc;

    border-radius: 8px;

    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);

}

.form-group {

    margin-bottom: 15px;

}
```



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```
label {  
    display: block;  
    margin-bottom: 4px;  
    font-weight: bold;  
}  
  
input {  
    width: 100%;  
    padding: 8px;  
    box-sizing: border-box;  
    border: 1px solid #ccc;  
    border-radius: 4px;  
    outline: none;  
}  
  
/* Highlight invalid input fields */  
input.invalid-input {  
    border: 2px solid #dc3545;  
    background-color: #fff8f8;  
}  
  
/* Style inline error text under inputs */  
.field-error {  
    color: #dc3545;  
    font-size: 13px;
```



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```
margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */
```

```
}
```

```
button {
```

```
width: 100%;

padding: 10px;

background-color: #007bff;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 16px;

margin-top: 10px;

}
```

```
button:hover {
```

```
background-color: #0056b3;

}
```

```
#result {
```

```
margin-top: 20px;

padding: 12px;

border-radius: 4px;
```



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```
display: none;

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}

</style>

</head>

<body>

<h2>Grading System</h2>

<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>

  <div class="form-group">

    <label for="studentName">Student Name:</label>

    <input type="text" id="studentName" placeholder="e.g. John Doe">

    <span class="field-error" id="nameError"></span>

  </div>

  <div class="form-group">

    <label for="math">Math Marks (0 - 100):</label>

    <input type="number" id="math" placeholder="e.g. 85">
```



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</div>

<div class="form-group">

<label for="science">Science Marks (0 - 100):</label>

<input type="number" id="science" placeholder="e.g. 90">

</div>

<div class="form-group">

<label for="english">English Marks (0 - 100):</label>

<input type="number" id="english" placeholder="e.g. 78">

</div>

<button type="submit">Calculate Overall Grade</button>

</form>

<div id="result"></div>

<script>

function calculateAverageGrade(event) {



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```
event.preventDefault();
```

```
// Track whether the form is completely valid
```

```
let isValid = true;
```

```
// Reset previous result box
```

```
const resultDiv = document.getElementById("result");
```

```
resultDiv.style.display = "none";
```

```
// -----
```

```
// Helper Function: Validate an individual field
```

```
// -----
```

```
function validateField(inputId, errorId, fieldName, isSubject = true) {
```

```
    const inputElement = document.getElementById(inputId);
```

```
    const errorElement = document.getElementById(errorId);
```

```
    const value = inputElement.value.trim();
```

```
// Clear previous styles/messages
```

```
inputElement.classList.remove("invalid-input");
```

```
errorElement.textContent = "";
```

```
// Rule 1: Empty text validation
```



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```
if (value === "") {  
    inputElement.classList.add("invalid-input");  
    errorElement.textContent = `Please enter ${fieldName}.`;   
    isValid = false;  
    return;  
}  
  
// Rule 2: Numeric & Range validation (for subject marks)  
if (isSubject) {  
    const score = parseFloat(value);  
    if (isNaN(score) || score < 0 || score > 100) {  
        inputElement.classList.add("invalid-input");  
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;   
        isValid = false;  
    }  
}  
  
// -----  
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY  
// -----  
  
validateField("studentName", "nameError", "student's name", false);
```



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```
validateField("math", "mathError", "Math marks", true);

validateField("science", "scienceError", "Science marks", true);

validateField("english", "englishError", "English marks", true);


// Stop calculation if any field failed validation
if (!isValid) return;


// -----

// 2. CALCULATE GRADE (Runs only if all fields are valid)
// -----

const math = parseFloat(document.getElementById("math").value);
const science = parseFloat(document.getElementById("science").value);
const english = parseFloat(document.getElementById("english").value);
const name = document.getElementById("studentName").value.trim();


const totalMarks = math + science + english;
const average = (totalMarks / 3).toFixed(2);


let grade = "";
let status = "";


if (average >= 90) {
```



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```
grade = "A+"; status = "Passed with Distinction";

} else if (average >= 80) {

    grade = "A"; status = "Passed";

} else if (average >= 70) {

    grade = "B"; status = "Passed";

} else if (average >= 60) {

    grade = "C"; status = "Passed";

} else if (average >= 50) {

    grade = "D"; status = "Passed";

} else {

    grade = "F"; status = "Failed";

}

// Display result summary

resultDiv.style.display = "block";

resultDiv.innerHTML = `

    <strong>Result Summary:</strong><br>

    Name: ${name}<br>

    Math: ${math} | Science: ${science} | English: ${english}<br>

    Total: ${totalMarks} / 300<br>

    Average: <strong>${average}%</strong><br>

    Overall Grade: <strong>${grade}</strong> (${status})
```



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```
`;  
}  
  
</script>
```

```
</body>
```

```
</html>
```

Output:

JS Practical - Multi-Subject Grading System

127.0.0.1:5500/Tejas_Bawane_24070521147/Experiment_3/grading_system.html

Grading System

Student Name:

Math Marks (0 - 100):

Science Marks (0 - 100):

English Marks (0 - 100):

[Calculate Overall Grade](#)

Result Summary:

Name: Tejas Bawane
Math: 95 | Science: 89 | English: 97
Total: 281 / 300
Average: 93.67%
Overall Grade: **A+** (Passed with Distinction)



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Case Study No.: 03

Title: Implement control structures and form validation; create a grading system based on user-entered marks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<html>
```

```
<head>
```

```
<title>JS Practical - Multi-Subject Grading System</title>
```

```
<style>
```

```
body {
```

```
    font-family: Arial, sans-serif;
```

```
    max-width: 480px;
```

```
    margin: 30px auto;
```

```
    padding: 20px;
```

```
    border: 1px solid #ccc;
```

```
    border-radius: 8px;
```

```
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
```

```
}
```

```
.form-group {
```



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margin-bottom: 15px;

}

label {

display: block;

margin-bottom: 4px;

font-weight: bold;

}

input {

width: 100%;

padding: 8px;

box-sizing: border-box;

border: 1px solid #ccc;

border-radius: 4px;

outline: none;

}

/* Highlight invalid input fields */

input.invalid-input {

border: 2px solid #dc3545;

background-color: #fff8f8;



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}

/* Style inline error text under inputs */

.field-error {

color: #dc3545;

font-size: 13px;

margin-top: 4px;

display: block;

min-height: 18px; /* Reserves space so layout doesn't jump */

}

button {

width: 100%;

padding: 10px;

background-color: #007bff;

color: white;

border: none;

border-radius: 4px;

cursor: pointer;

font-size: 16px;

margin-top: 10px;



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}

button:hover {

background-color: #0056b3;

}

#result {

margin-top: 20px;

padding: 12px;

border-radius: 4px;

display: none;

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}

</style>

</head>

<body>

<h2>Grading System</h2>



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```
<form id="gradingForm" onsubmit="calculateAverageGrade(event)" novalidate>
```

```
<div class="form-group">
```

```
<label for="studentName">Student Name:</label>
```

```
<input type="text" id="studentName" placeholder="e.g. John Doe">
```

```
<span class="field-error" id="nameError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="math">Math Marks (0 - 100):</label>
```

```
<input type="number" id="math" placeholder="e.g. 85">
```

```
<span class="field-error" id="mathError"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="science">Science Marks (0 - 100):</label>
```

```
<input type="number" id="science" placeholder="e.g. 90">
```

```
<span class="field-error" id="scienceError"></span>
```

```
</div>
```



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```
<div class="form-group">
```

```
<label for="english">English Marks (0 - 100):</label>
```

```
<input type="number" id="english" placeholder="e.g. 78">
```

```
<span class="field-error" id="englishError"></span>
```

```
</div>
```

```
<button type="submit">Calculate Overall Grade</button>
```

```
</form>
```

```
<div id="result"></div>
```

```
<script>
```

```
function calculateAverageGrade(event) {
```

```
    event.preventDefault();
```

```
    // Track whether the form is completely valid
```

```
    let isValid = true;
```



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```
// Reset previous result box

const resultDiv = document.getElementById("result");

resultDiv.style.display = "none";


// -----

// Helper Function: Validate an individual field

// -----

function validateField(inputId, errorId, fieldName, isSubject = true) {

    const inputElement = document.getElementById(inputId);

    const errorElement = document.getElementById(errorId);

    const value = inputElement.value.trim();

    // Clear previous styles/messages

    inputElement.classList.remove("invalid-input");

    errorElement.textContent = "";

    // Rule 1: Empty text validation

    if (value === "") {

        inputElement.classList.add("invalid-input");
```



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```
errorElement.textContent = `Please enter ${fieldName}.`;
```

```
isValid = false;
```

```
return;
```

```
}
```

```
// Rule 2: Numeric & Range validation (for subject marks)
```

```
if (isSubject) {
```

```
    const score = parseFloat(value);
```

```
    if (isNaN(score) || score < 0 || score > 100) {
```

```
        inputElement.classList.add("invalid-input");
```

```
        errorElement.textContent = `Enter valid ${fieldName} (0–100).`;
```

```
        isValid = false;
```

```
    }
```

```
}
```

```
}
```

```
// -----
```

```
// 1. RUN VALIDATIONS FOR ALL FIELDS INDEPENDENTLY
```

```
// -----
```



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```
validateField("studentName", "nameError", "student's name", false);
```

```
validateField("math", "mathError", "Math marks", true);
```

```
validateField("science", "scienceError", "Science marks", true);
```

```
validateField("english", "englishError", "English marks", true);
```

```
// Stop calculation if any field failed validation
```

```
if (!isValid) return;
```

```
// -----
```

```
// 2. CALCULATE GRADE (Runs only if all fields are valid)
```

```
// -----
```

```
const math = parseFloat(document.getElementById("math").value);
```

```
const science = parseFloat(document.getElementById("science").value);
```

```
const english = parseFloat(document.getElementById("english").value);
```

```
const name = document.getElementById("studentName").value.trim();
```

```
const totalMarks = math + science + english;
```

```
const average = (totalMarks / 3).toFixed(2);
```



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```
let grade = "";

let status = "";

let scholarship = "";

if (average >= 95) {

    grade = "A+"; status = "Passed with Distinction"; scholarship = "Eligible for Platinum
Scholarship";

} else if (average >= 85) {

    grade = "A"; status = "Passed"; scholarship = "Eligible for Gold Scholarship";

} else if (average >= 70) {

    grade = "B"; status = "Passed"; scholarship = "Eligible for Silver Scholarship";

} else if (average >= 50) {

    grade = "C"; status = "Passed"; scholarship = "Eligible for Bronze Scholarship";

} else if (average < 50) {

    grade = "D"; status = "Passed"; scholarship = "No Scholarship";

} else {

    grade = "F"; status = "Failed";

}
```



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// Display result summary

```
resultDiv.style.display = "block";
```

```
resultDiv.innerHTML = `
```

```
    <strong>Result Summary:</strong><br>
```

```
    Name: ${name}<br>
```

```
    Math: ${math} | Science: ${science} | English: ${english}<br>
```

```
    Total: ${totalMarks} / 300<br>
```

```
    Average: <strong>${average}%</strong><br>
```

```
    Overall Grade: <strong>${grade}</strong> (${status}) ${scholarship}
```

```
`;
```

```
}
```

```
</script>
```

```
</body>
```

```
</html>
```

Output:



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JS Practical - Multi-Subject Grading

127.0.0.1:5500/Tejas_Bawane_24070521147/Case%20Study%203/scholarship_grading_system.html

Grading System

Student Name:

Math Marks (0 - 100):

Science Marks (0 - 100):

English Marks (0 - 100):

[Calculate Overall Grade](#)

Result Summary:

Name: Tejas Bawane
Math: 95 | Science: 89 | English: 97
Total: 281 / 300
Average: 93.67%
Overall Grade: **A (Passed)** Eligible for Gold Scholarship



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Experiment No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
let welcome = function() {  
    console.log("Welcome to the Palindrome Checker!");  
};  
  
welcome();  
  
// Global variable  
let language = "JavaScript";  
  
// Main function to check if a string or number is a palindrome  
function isPalindrome(str) {  
    // 1. Accept numbers by converting them to strings  
    if (typeof str === "number") {  
        str = String(str);  
    }  
  
    // 2. Validate input type (reject null, undefined, objects, arrays, etc.)
```



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```
if (typeof str !== "string") {  
    throw new TypeError(`Expected a string or number, but received ${typeof str}`);  
}  
  
// 3. Check for empty string  
if (str.trim().length === 0) {  
    throw new Error("Input string cannot be empty.");  
}  
  
// Clean the string: remove non-alphanumeric characters and convert to lowercase  
let cleaned = str.toLowerCase().replace(/[^a-z0-9]/g, "");  
  
// Handle edge case where input had only punctuation/spaces (e.g., "!!!")  
if (cleaned.length === 0) {  
    throw new Error("Input must contain at least one alphanumeric character.");  
}  
  
// Reverse and check  
let reversed = cleaned.split("").reverse().join("");  
return cleaned === reversed;  
}
```



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// Helper function with try...catch error handling

```
function testPalindrome(phrase) {
```

```
  try {
```

```
    let result = isPalindrome(phrase);
```

```
    console.log(` SUCCESS: "${phrase}" -> ${result ? "Is a palindrome!" : "Not a  
palindrome."}`);
```

```
  } catch (error) {
```

```
    // Handle caught errors safely without stopping execution
```

```
    console.error(` ERROR [${error.name}]: Failed to check input (${phrase}) ->  
${error.message}`);
```

```
  }
```

```
}
```

// --- Testing Examples ---

```
console.log(`\nRunning checks using ${language}:\n` + "=".repeat(45));
```

// Valid Test Cases (Strings & Numbers)

```
testPalindrome("Hello World");
```

```
testPalindrome("Tejas Bawane");
```

```
testPalindrome("24070521147");
```

```
testPalindrome(12321);           // NOW WORKS: Passed as a Number directly!
```

```
testPalindrome(21212);           // NOW WORKS: Evaluates as "Not a palindrome"
```



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// Error Handling Test Cases (Triggers Catch Block)

testPalindrome(""); // Error: Empty string

testPalindrome(null); // TypeError: null passed

testPalindrome(" "); // Error: Whitespace only

testPalindrome("!!!?"); // Error: No alphanumeric characters

Output:

```
PS E:\JS lab> node "e:\JS lab\Tejas_Bawane_24070521147\Experiment_4\palindrome_checker.js"
Welcome to the Palindrome Checker!

Running checks using JavaScript:
=====
SUCCESS: "Hello World" -> Not a palindrome.
SUCCESS: "Tejas Bawane" -> Not a palindrome.
SUCCESS: "24070521147" -> Not a palindrome.
SUCCESS: "12321" -> Is a palindrome!
SUCCESS: "21212" -> Is a palindrome!
ERROR [Error]: Failed to check input () -> Input string cannot be empty.
ERROR [TypeError]: Failed to check input (null) -> Expected a string or number, but received object
ERROR [Error]: Failed to check input ( ) -> Input string cannot be empty.
ERROR [Error]: Failed to check input (!!!?) -> Input must contain at least one alphanumeric character.
```



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Case Study No.: 04

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
const readline = require('readline');
```

```
// 1. FUNCTION DECLARATION (Global Scope)
```

```
// Utility function to reverse the entered PIN
```

```
function reversePin(pin) {  
    const pinStr = String(pin);  
    return pinStr.split("").reverse().join("");  
}
```

```
// Helper function to ask a question in the terminal using built-in Node readline
```

```
const askQuestion = (query) => {  
    const rl = readline.createInterface({  
        input: process.stdin,  
        output: process.stdout  
    });  
    return new Promise(resolve => rl.question(query, answer => {
```



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```
rl.close();

resolve(answer);

});

};

// 2. CLOSURE & SCOPE DEMONSTRATION

// ATM Session Generator returning an Arrow Function

const createATMSession = (accountHolder) => {

  // Outer Lexical Scope variable

  const user = accountHolder;

  let attempts = 0;

  // 3. ARROW FUNCTION (Closure over `user` and `attempts`)

  const verifyPinInteractive = async () => {

    attempts++;

    // Dynamic prompt to ask the user for PIN input

    const enteredPin = await askQuestion(`[ATM System] ${user}, please enter any PIN: `);
```



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```
if (!enteredPin || enteredPin.trim() === "") {  
    return `Transaction cancelled by ${user}.`;  
}  
  
const strEnteredPin = String(enteredPin.trim());  
const reversed = reversePin(strEnteredPin); // Calls global Function Declaration  
  
console.log(`\n--- Transaction Attempt #${attempts} for ${user} ---`);  
console.log(`Entered PIN : ${strEnteredPin}`);  
console.log(`Reversed PIN: ${reversed}`);  
  
// Check if the entered PIN is a Palindrome  
const isPalindrome = (strEnteredPin === reversed);  
  
// Access is ALWAYS granted, but security message depends on Palindrome check  
if (isPalindrome) {  
    return `ACCESS GRANTED!\n SPECIAL SECURITY ALERT: PIN "${strEnteredPin}"  
is a Palindrome! Please consider changing your password for better security.`;  
} else {  
    return `ACCESS GRANTED! Welcome, ${user}.`;  
}  
};
```



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```
return verifyPinInteractive;

};

// 4. ATM SYSTEM EXECUTION

(async () => {

    const johnSession = createATMSession("John Doe");

    // Prompts user in the integrated terminal

    const result = await johnSession();

    console.log("\n" + result);

})();
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS E:\JS lab> node "e:\JS lab\Tejas_Bawane_24070521147\Case Study 4\ATM_pin_verification.js"
[ATM System] John Doe, please enter any PIN: 1147

--- Transaction Attempt #1 for John Doe ---
Entered PIN : 1147
Reversed PIN: 7411

ACCESS GRANTED! Welcome, John Doe.
❖ PS E:\JS lab> 
```



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Experiment No.: 05

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
let welcome = function() {  
    console.log("Welcome to the Palindrome Checker!");  
};  
  
welcome();  
  
// Global variable  
let language = "JavaScript";  
  
// Main function to check if a string or number is a palindrome  
function isPalindrome(str) {  
    // 1. Accept numbers by converting them to strings  
    if (typeof str === "number") {  
        str = String(str);  
    }  
  
    // 2. Validate input type (reject null, undefined, objects, arrays, etc.)  
    if (typeof str !== "string") {
```



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```
    throw new TypeError(`Expected a string or number, but received ${typeof str}`);
  }

  // 3. Check for empty string
  if (str.trim().length === 0) {
    throw new Error("Input string cannot be empty.");
  }

  // Clean the string: remove non-alphanumeric characters and convert to lowercase
  let cleaned = str.toLowerCase().replace(/[^a-z0-9]/g, "");

  // Handle edge case where input had only punctuation/spaces (e.g., "!!!")
  if (cleaned.length === 0) {
    throw new Error("Input must contain at least one alphanumeric character.");
  }

  // Reverse and check
  let reversed = cleaned.split("").reverse().join("");
  return cleaned === reversed;
}

// Helper function with try...catch error handling
```



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```
function testPalindrome(phrase) {  
  
  try {  
  
    let result = isPalindrome(phrase);  
  
    console.log(` SUCCESS: "${phrase}" -> ${result ? "Is a palindrome!" : "Not a  
palindrome."}`);  
  
    } catch (error) {  
  
      // Handle caught errors safely without stopping execution  
  
      console.error(` ERROR [${error.name}]: Failed to check input (${phrase}) ->  
${error.message}`);  
  
    }  
  
  }  
  
  // --- Testing Examples ---  
  
  console.log(`\nRunning checks using ${language}:\n` + "=".repeat(45));  
  
  
  // Valid Test Cases (Strings & Numbers)  
  
  testPalindrome("Hello World");  
  
  testPalindrome("Aishwarya Brahmkar");  
  
  testPalindrome("24070521212");  
  
  testPalindrome(12321);           // NOW WORKS: Passed as a Number directly!  
  
  testPalindrome(21212);           // NOW WORKS: Evaluates as "Not a palindrome"  
  
  
  // Error Handling Test Cases (Triggers Catch Block)
```



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```
testPalindrome("");           // Error: Empty string
testPalindrome(null);         // TypeError: null passed
testPalindrome(" ");          // Error: Whitespace only
testPalindrome("!!!?");       // Error: No alphanumeric characters
```

Output:

```
[Running] node "e:\JS lab\Experiment_4\palindrome_checker.js"
Welcome to the Palindrome Checker!
ERROR [Error]: Failed to check input () -> Input string cannot be empty.
ERROR [TypeError]: Failed to check input (null) -> Expected a string or number, but received object
ERROR [Error]: Failed to check input ( ) -> Input string cannot be empty.

Running checks using JavaScript:
=====
SUCCESS: "Hello World" -> Not a palindrome.
SUCCESS: "Aishwarya Brahmkar" -> Not a palindrome.
SUCCESS: "24070521212" -> Not a palindrome.
SUCCESS: "12321" -> Is a palindrome!
SUCCESS: "21212" -> Is a palindrome!
ERROR [Error]: Failed to check input (!!!?) -> Input must contain at least one alphanumeric character.

[Done] exited with code=0 in 0.158 seconds
```



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Case Study No.: 05

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
const readline = require('readline');
```

```
// 1. FUNCTION DECLARATION (Global Scope)
```

```
// Utility function to reverse the entered PIN
```

```
function reversePin(pin) {  
    const pinStr = String(pin);  
    return pinStr.split("").reverse().join("");  
}
```

```
// Helper function to ask a question in the terminal using built-in Node readline
```

```
const askQuestion = (query) => {  
    const rl = readline.createInterface({  
        input: process.stdin,  
        output: process.stdout  
    });  
    return new Promise(resolve => rl.question(query, answer => {  
        rl.close();
```



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```
    resolve(answer);

  });

};

// 2. CLOSURE & SCOPE DEMONSTRATION

// ATM Session Generator returning an Arrow Function

const createATMSession = (accountHolder) => {

  // Outer Lexical Scope variable

  const user = accountHolder;

  let attempts = 0;

  // 3. ARROW FUNCTION (Closure over `user` and `attempts`)

  const verifyPinInteractive = async () => {

    attempts++;

    // Dynamic prompt to ask the user for PIN input

    const enteredPin = await askQuestion(`[ATM System] ${user}, please enter any PIN: `);
```



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```
if (!enteredPin || enteredPin.trim() === "") {  
    return `Transaction cancelled by ${user}.`;  
}  
  
const strEnteredPin = String(enteredPin.trim());  
const reversed = reversePin(strEnteredPin); // Calls global Function Declaration  
  
console.log(`\n--- Transaction Attempt #${attempts} for ${user} ---`);  
console.log(`Entered PIN : ${strEnteredPin}`);  
console.log(`Reversed PIN: ${reversed}`);  
  
// Check if the entered PIN is a Palindrome  
const isPalindrome = (strEnteredPin === reversed);  
  
// Access is ALWAYS granted, but security message depends on Palindrome check  
if (isPalindrome) {  
    return `ACCESS GRANTED!\n SPECIAL SECURITY ALERT: PIN "${strEnteredPin}"  
is a Palindrome! Please consider changing your password for better security.`;  
} else {  
    return `ACCESS GRANTED! Welcome, ${user}.`;  
}  
};
```



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```
return verifyPinInteractive;  
  
};  
  
// 4. ATM SYSTEM EXECUTION  
  
(async () => {  
  
    const johnSession = createATMSession("John Doe");  
  
    // Prompts user in the integrated terminal  
  
    const result = await johnSession();  
  
    console.log("\n" + result);  
  
})();
```

Output:

```
PS E:\JS lab> node "e:\JS lab\Aishwarya_Brahmankar_24070521212\Case Study 4\ATM_pin_verification.js"  
[ATM System] John Doe, please enter any PIN: 1221  
  
--- Transaction Attempt #1 for John Doe ---  
Entered PIN : 1221  
Reversed PIN: 1221  
  
ACCESS GRANTED!  
SPECIAL SECURITY ALERT: PIN "1221" is a Palindrome! Please consider changing your password for better security.
```



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Experiment No.: 06

Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
let welcome = function() {  
    console.log("Welcome to the Palindrome Checker!");  
};  
  
welcome();  
  
// Global variable  
let language = "JavaScript";  
  
// Main function to check if a string or number is a palindrome  
function isPalindrome(str) {  
    // 1. Accept numbers by converting them to strings  
    if (typeof str === "number") {  
        str = String(str);  
    }  
  
    // 2. Validate input type (reject null, undefined, objects, arrays, etc.)  
    if (typeof str !== "string") {
```



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```
        throw new TypeError(`Expected a string or number, but received ${typeof str}`);
    }

    // 3. Check for empty string
    if (str.trim().length === 0) {
        throw new Error("Input string cannot be empty.");
    }

    // Clean the string: remove non-alphanumeric characters and convert to lowercase
    let cleaned = str.toLowerCase().replace(/[^a-z0-9]/g, "");

    // Handle edge case where input had only punctuation/spaces (e.g., "!!!")
    if (cleaned.length === 0) {
        throw new Error("Input must contain at least one alphanumeric character.");
    }

    // Reverse and check
    let reversed = cleaned.split("").reverse().join("");
    return cleaned === reversed;
}

// Helper function with try...catch error handling
```



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```
function testPalindrome(phrase) {  
  
  try {  
  
    let result = isPalindrome(phrase);  
  
    console.log(` SUCCESS: "${phrase}" -> ${result ? "Is a palindrome!" : "Not a  
palindrome."}`);  
  
    } catch (error) {  
  
      // Handle caught errors safely without stopping execution  
  
      console.error(` ERROR [${error.name}]: Failed to check input (${phrase}) ->  
${error.message}`);  
  
    }  
  
  }  
  
  // --- Testing Examples ---  
  
  console.log(`\nRunning checks using ${language}:\n` + "=".repeat(45));  
  
  
  // Valid Test Cases (Strings & Numbers)  
  
  testPalindrome("Hello World");  
  
  testPalindrome("Aishwarya Brahmkar");  
  
  testPalindrome("24070521212");  
  
  testPalindrome(12321);           // NOW WORKS: Passed as a Number directly!  
  
  testPalindrome(21212);           // NOW WORKS: Evaluates as "Not a palindrome"  
  
  
  // Error Handling Test Cases (Triggers Catch Block)
```



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```
testPalindrome("");           // Error: Empty string
testPalindrome(null);         // TypeError: null passed
testPalindrome(" ");          // Error: Whitespace only
testPalindrome("!!!?");        // Error: No alphanumeric characters
```

Output:

```
Occurrences of 'email': 0
PS E:\Tejas_Bawane_24070521147> node "e:\Tejas_Bawane_24070521147\Experiments\Experiment_6\data_extraction.js"
Valid email address true
Extracted Emails: [ 'tejas.bawane.batch2024@sitnagpur.siu.edu.in.' ]
Uppercase THIS IS TEJAS BAWANE, CONTACT ME AT TEJAS.BAWANE.BATCH2024@SITNAGPUR.SIU.EDU.IN. WELCOME TO OUR WEBSITE!
Lowercase this is tejas bawane, contact me at tejas.bawane.batch2024@sitnagpur.siu.edu.in. welcome to our website!
Contains 'Welcome': true
Text Length 104
Number of Words: 12
Occurrences of 'email': 0
❖ PS E:\Tejas_Bawane_24070521147>
```



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Case Study No.: 06

Title: Build a palindrome checker to check if the reversed pin is palindrome and ask the user to change the password to a stronger one.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

Count Vowels :

```
function countVowels(paragraph) {  
    const vowels = "aeiouAEIOU";  
    let count = 0;  
  
    for (let char of paragraph) {  
        if (vowels.includes(char)) {  
            count++;  
        }  
    }  
  
    return count;  
}
```

// Test

```
const para = "JavaScript is a versatile programming language used for web development.";  
console.log("Total Vowels:", countVowels(para));
```



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Output:

```
PS E:\Aishwarya_Brahmankar_24070521212> node "e:\Aishwarya_Brahmankar_24070521212\Case Study\Case Study 6\count_vowels.js"
Total Vowels: 24
PS E:\Aishwarya_Brahmankar_24070521212>
```

Reverse String :

```
function reverseString(str) {

    // 1. split("") converts string into array of characters

    // 2. reverse() reverses array in-place

    // 3. join("") combines array elements back into a string

    return str.split("").reverse().join("");

}
```

```
// Test

const inputStr = "Hello World";

console.log("Original:", inputStr);

console.log("Reversed:", reverseString(inputStr));
```

Output :

```
PS E:\Aishwarya_Brahmankar_24070521212> node "e:\Aishwarya_Brahmankar_24070521212\Case Study\Case Study 6\reverse_string.js"
Original: Hello World
Reversed: dlroW olleH
PS E:\Aishwarya_Brahmankar_24070521212>
```



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String Basics :

// A. Basics of Strings

```
let text = "JavaScript Programming";
```

```
console.log(text.length);    // 22 (Property: returns total characters)
```

```
console.log(text.charAt(0));  // "J" (Gets character at specified index)
```

```
console.log(text.toLowerCase()); // "javascript programming"
```

```
console.log(text.toUpperCase()); // "JAVASCRIPT PROGRAMMING"
```

// B. Core String Methods

// 1. substring(start, end) - Extracts characters between start and end (excluding end)

```
let sub = text.substring(0, 10);
```

```
console.log(sub); // "JavaScript"
```

// 2. indexOf(searchValue) - Returns first index of matching substring (-1 if not found)

```
let position = text.indexOf("Prog");
```

```
console.log(position); // 11
```

// 3. split(separator) - Splits string into an array of substrings

```
let words = text.split(" ");
```

```
console.log(words); // ["JavaScript", "Programming"]
```



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// 4. replace(searchValue, newValue) - Replaces the first matching value

```
let newText = text.replace("JavaScript", "JS");
```

```
console.log(newText); // "JS Programming"
```

Output :

```
PS E:\Aishwarya_Brahmankar_24070521212> node "e:\Aishwarya_Brahmankar_24070521212\Case Study\Case Study 6\string_basics.js"
22
J
javascript programming
JAVASCRIPT PROGRAMMING
JavaScript
11
[ 'JavaScript', 'Programming' ]
JS Programming
PS E:\Aishwarya_Brahmankar_24070521212>
```



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Experiment No.: 07

Title: Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>To-Do List App</title>

  <style>

    * {

      box-sizing: border-box;

      font-family: Arial, sans-serif;

    }

    body {

      background-color: #d1d8e0;

      display: flex;

      justify-content: center;

      align-items: flex-start;
```



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```
padding-top: 40px;

min-height: 100vh;

margin: 0;

}

.todo-container {

    background-color: #f1f2f6;

    width: 550px;

    padding: 30px;

    border-radius: 8px;

    box-shadow: 0 4px 15px rgba(0,0,0,0.15);

}

h1 {

    text-align: center;

    color: #333;

    margin-top: 0;

    margin-bottom: 25px;

    font-size: 2.2rem;

}

.input-section {
```



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```
display: flex;

gap: 10px;

margin-bottom: 25px;
}

#taskInput {

  flex: 1;

  padding: 12px 15px;

  font-size: 1rem;

  border: 1px solid #a4b0be;

  border-radius: 4px;

  outline: none;
}

#taskInput:focus {

  border-color: #2ed573;
}

button {

  cursor: pointer;

  border: none;

  border-radius: 4px;
```



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```
font-weight: bold;

font-size: 1rem;

transition: opacity 0.2s;

}

button:hover {

  opacity: 0.9;

}

.add-btn {

  background-color: #2ed573;

  color: white;

  padding: 0 20px;

}

.task-list {

  list-style: none;

  padding: 0;

  margin: 0 0 20px 0;

}

.task-item {
```



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```
background-color: #e4e7eb;

display: flex;

align-items: center;

padding: 12px 15px;

border-radius: 6px;

margin-bottom: 12px;

gap: 12px;
}

.task-item.completed .task-text {

  text-decoration: line-through;

  color: #747d8c;
}

.task-checkbox {

  width: 16px;

  height: 16px;

  cursor: pointer;
}

.task-text {

  flex: 1;
}
```



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```
font-size: 1.05rem;

color: #2f3542;

word-break: break-word;
}

.edit-input {
  flex: 1;
  padding: 6px 10px;
  font-size: 1rem;
  border: 1px solid #ffa502;
  border-radius: 4px;
}

.edit-btn {
  background-color: #ffa502;
  color: white;
  padding: 8px 16px;
}

.delete-btn {
  background-color: #ff4757;
  color: white;
```



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```
padding: 8px 16px;

}

.task-counter {

  text-align: center;

  font-weight: bold;

  color: #2f3542;

  font-size: 1.1rem;

  margin-top: 20px;

}

</style>

</head>

<body>

  <div class="todo-container">

    <h1>To-Do List</h1>

    <div class="input-section">

      <input type="text" id="taskInput" placeholder="Enter a task">

      <button class="add-btn" id="addTaskBtn">Add</button>

    </div>
```



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<!-- The list is now empty by default -->

<ul class="task-list" id="taskList">

<div class="task-counter" id="taskCounter">Total Tasks: 0</div>

</div>

<script>

// DOM Elements

const taskInput = document.getElementById('taskInput');

const addTaskBtn = document.getElementById('addTaskBtn');

const taskList = document.getElementById('taskList');

const taskCounter = document.getElementById('taskCounter');

// Function to update the total task counter

function updateCounter() {

const totalTasks = taskList.children.length;

taskCounter.textContent = `Total Tasks: \${totalTasks}`;

}

// Function to add a new task item

function addTask() {

const taskValue = taskInput.value.trim();



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```
if (taskValue === "") {  
    alert('Please enter a valid task!');  
    return;  
}  
  
// Create list element  
const li = document.createElement('li');  
li.className = 'task-item';  
  
// Create and append dynamic internal DOM structure  
li.innerHTML = `  
    <input type="checkbox" class="task-checkbox">  
    <span class="task-text">${taskValue}</span>  
    <button class="edit-btn">Edit</button>  
    <button class="delete-btn">Delete</button>  
`;  
  
taskList.appendChild(li);  
taskInput.value = ""; // Reset input field  
updateCounter();  
}
```



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// Event listener for the Add button

```
addTaskBtn.addEventListener('click', addTask);
```

// Allow pressing 'Enter' key to add a task

```
taskInput.addEventListener('keypress', function(e) {  
    if (e.key === 'Enter') {  
        addTask();  
    }  
});
```

// Event Delegation for operations inside the list (Edit, Save, Delete, Toggle Checkbox)

```
taskList.addEventListener('click', function(e) {  
    const target = e.target;  
    const parentLI = target.parentElement; // DOM Traversal to parent element
```

// 1. DELETE FEATURE

```
if (target.classList.contains('delete-btn')) {  
    parentLI.remove();  
    updateCounter();  
}
```

// 2. EDIT/SAVE FEATURE



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```
else if (target.classList.contains('edit-btn')) {

    const taskTextSpan = parentLI.querySelector('.task-text');

    if (target.textContent === 'Edit') {

        // Switch to edit mode: replace text span with an input text box

        const currentText = taskTextSpan.textContent;

        const inputField = document.createElement('input');

        inputField.type = 'text';

        inputField.className = 'edit-input';

        inputField.value = currentText;

        parentLI.replaceChild(inputField, taskTextSpan); // DOM update replacement

        target.textContent = 'Save';

        target.style.backgroundColor = '#2ed573'; // Change button color to green on save
mode

    } else {

        // Save mode processing

        const editInput = parentLI.querySelector('.edit-input');

        const updatedText = editInput.value.trim();

        if (updatedText === "") {

            alert('Task cannot be empty!');

            return;
```



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```
}
```

```
const newSpan = document.createElement('span');
```

```
newSpan.className = 'task-text';
```

```
newSpan.textContent = updatedText;
```

```
parentLI.replaceChild(newSpan, editInput); // DOM update replacement
```

```
target.textContent = 'Edit';
```

```
target.style.backgroundColor = '#ffa502'; // Restore button color to orange
```

```
}
```

```
}
```

```
// 3. TOGGLE CHECKBOX (COMPLETION) STATUS
```

```
else if (target.classList.contains('task-checkbox')) {
```

```
  if (target.checked) {
```

```
    parentLI.classList.add('completed');
```

```
  } else {
```

```
    parentLI.classList.remove('completed');
```

```
  }
```

```
}
```

```
});
```



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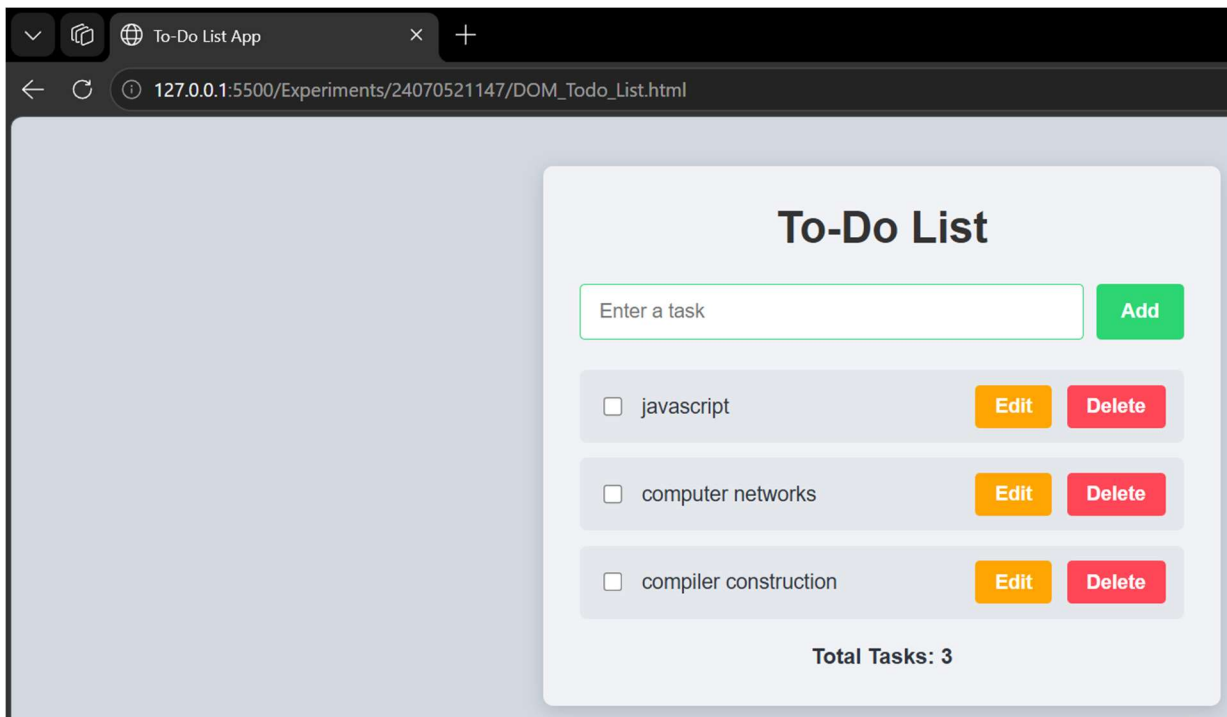
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```
// Initial counter setup  
  
updateCounter();  
  
</script>  
  
</body>  
  
</html>
```

Output:





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Case Study No.: 07

Title: To demonstrate the Events Handling in Javascript

- a) Access and manipulate form elements using Javascript.
- b) Validate user input (e.g, email, empty fields)
- c) Use various form events (submit, change, focus)

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Form Handling with JavaScript</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      margin: 30px;

    }

    .form-container {

      border: 1px solid #000;
```



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```
width: 450px;

padding: 20px;

box-sizing: border-box;

}
```

```
.mandatory-note {

  color: red;

  font-size: 13px;

  margin-bottom: 15px;

}
```

```
.form-row {

  display: flex;

  align-items: center;

  margin-bottom: 10px;

}
```

```
.form-row label {

  width: 110px;

  font-size: 14px;

}
```



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```
.form-row input[type="text"],  
.form-row input[type="email"],  
.form-row input[type="url"],  
.form-row input[type="password"],  
.form-row select {  
    padding: 3px;  
    border: 1px solid #777;  
}  
  
.input-field {  
    width: 180px;  
}  
  
.asterisk {  
    color: red;  
    margin-left: 4px;  
}  
  
.date-selects select {  
    margin-right: 2px;  
}
```



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```
.checkbox-row {  
    display: flex;  
    align-items: center;  
    margin-top: 15px;  
    margin-bottom: 15px;  
}
```

```
.checkbox-row label {  
    margin-left: 5px;  
    font-size: 14px;  
}
```

```
.submit-btn {  
    padding: 4px 12px;  
    font-weight: bold;  
    cursor: pointer;  
}
```

```
/* Input highlighting on focus event */
```

```
input:focus, select:focus {  
    background-color: #f0f8ff;  
    outline: 1px solid #0056b3;
```



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}

</style>

</head>

<body>

<div class="form-container">

<div class="mandatory-note">*Mandatory to fill</div>

<form id="signupForm" novalidate>

<div class="form-row">

<label for="firstname">Firstname:</label>

<input type="text" id="firstname" class="input-field">

*

</div>

<div class="form-row">

<label for="lastname">Lastname:</label>

<input type="text" id="lastname" class="input-field">

</div>

<div class="form-row">



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<label>Birthday:</label>

<div class="date-selects">

<select id="day">

<option value="">Day</option>

</select>

<select id="month">

<option value="">Month</option>

<option value="1">Jan</option>

<option value="2">Feb</option>

<option value="3">Mar</option>

<option value="4">Apr</option>

<option value="5">May</option>

<option value="6">Jun</option>

<option value="7">Jul</option>

<option value="8">Aug</option>

<option value="9">Sep</option>

<option value="10">Oct</option>

<option value="11">Nov</option>

<option value="12">Dec</option>

</select>

<select id="year">

<option value="">Year</option>



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</select>

</div>

*

</div>

<div class="form-row">

<label for="username">Username:</label>

<input type="text" id="username" class="input-field">

*

</div>

<div class="form-row">

<label for="email">E-mail:</label>

<input type="email" id="email" class="input-field">

*

</div>

<div class="form-row">

<label for="website">Website:</label>

<input type="url" id="website" class="input-field">

*

</div>



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```
<div class="form-row">
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="repassword">Re-password:</label>
```

```
<input type="password" id="repassword" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="checkbox-row">
```

```
<input type="checkbox" id="terms">
```

```
<label for="terms">I agree to the terms & conditions.</label>
```

```
</div>
```

```
<button type="submit" class="submit-btn">Sign up</button>
```

```
</form>
```

```
</div>
```



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<script>

// Populate Days (1 to 31)

const daySelect = document.getElementById('day');

for (let i = 1; i <= 31; i++) {

const opt = document.createElement('option');

opt.value = i;

opt.textContent = i;

daySelect.appendChild(opt);

}

// Populate Years (1950 to 2026)

const yearSelect = document.getElementById('year');

for (let i = 2026; i >= 1950; i--) {

const opt = document.createElement('option');

opt.value = i;

opt.textContent = i;

yearSelect.appendChild(opt);

}

// Event Listener: Change event demonstration

document.getElementById('month').addEventListener('change', function() {

console.log('Month changed to: ' + this.value);



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```
});
```

```
// Event Listener: Submit event and field validations
```

```
document.getElementById('signupForm').addEventListener('submit', function(e) {  
    e.preventDefault();
```

```
    const firstname = document.getElementById('firstname').value.trim();
```

```
    const day = document.getElementById('day').value;
```

```
    const month = document.getElementById('month').value;
```

```
    const year = document.getElementById('year').value;
```

```
    const username = document.getElementById('username').value.trim();
```

```
    const email = document.getElementById('email').value.trim();
```

```
    const website = document.getElementById('website').value.trim();
```

```
    const password = document.getElementById('password').value;
```

```
    const repassword = document.getElementById('repassword').value;
```

```
    const terms = document.getElementById('terms').checked;
```

```
// Check empty fields
```

```
if (!firstname) {
```

```
    alert('Firstname is required.');
```

```
    return;
```

```
}
```



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```
if (!day || !month || !year) {
```

```
    alert('Please select a complete birthday.');
```

```
    return;
```

```
}
```

```
if (!username) {
```

```
    alert('Username is required.');
```

```
    return;
```

```
}
```

```
// Email validation
```

```
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (!emailRegex.test(email)) {
```

```
    alert('Please enter a valid email address.');
```

```
    return;
```

```
}
```

```
// Website validation
```

```
const urlRegex = /^(https?:\/\/)?([\w\d-]+\.)+[\w-]+(\.*)?$/i;
```

```
if (!urlRegex.test(website)) {
```

```
    alert('Please enter a valid website URL.');
```



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```
return;

}

// Password validations

if (!password) {

    alert('Password is required.');
```

return;

}

if (password !== repassword) {

alert('Passwords do not match.');

return;

}

// Terms & Conditions checkbox check

if (!terms) {

alert('You must agree to the terms & conditions.');

return;

}

alert('Form submitted successfully!');

});



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</script>

</body>

</html>

Output:

Form Handling with JavaScript

127.0.0.1:5500/Case%20Study/24070521212/login_form.html

***Mandatory to fill**

Firstname: *

Lastname:

Birthday: *

Username: *

E-mail: *

Website: *

Password: *

Re-password: *

☒ I agree to the terms & conditions.



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Experiment No.: 08

Title: Access and validate form fields using events; design a gym admission form with live input checks.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Gym Membership Form</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      background-color: #f4f4f9;

      display: flex;

      justify-content: center;

      align-items: center;

      min-height: 100vh;

      margin: 0;

      padding: 20px 0;

    }
```



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```
.form-container {  
  background-color: #ffffff;  
  padding: 30px;  
  border-radius: 8px;  
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);  
  width: 100%;  
  max-width: 420px;  
}
```

```
.form-container h2 {  
  margin-bottom: 20px;  
  color: #333;  
  text-align: center;  
}
```

```
.form-group {  
  margin-bottom: 15px;  
}
```

```
.form-group label {  
  display: block;
```



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```
margin-bottom: 5px;  
  
font-weight: bold;  
  
color: #555;  
  
}
```

```
.form-group input,  
.form-group select {  
    width: 100%;  
    padding: 10px;  
    border: 1px solid #ccc;  
    border-radius: 4px;  
    box-sizing: border-box;  
    font-size: 0.95rem;  
  
}
```

```
.error-message {  
    color: #e74c3c;  
    font-size: 0.85rem;  
    margin-top: 4px;  
    display: block;  
    min-height: 18px;  
  
}
```



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```
button {  
  width: 100%;  
  padding: 12px;  
  background-color: #28a745;  
  color: white;  
  border: none;  
  border-radius: 4px;  
  font-size: 1rem;  
  font-weight: bold;  
  cursor: pointer;  
  margin-top: 10px;  
}
```

```
button:hover {  
  background-color: #218838;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="form-container">
```



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<h2>Gym Membership Form</h2>

<form id="gymForm">

<div class="form-group">

<label for="name">Full Name</label>

<input type="text" id="name" placeholder="Enter full name">

</div>

<div class="form-group">

<label for="email">Email Address</label>

<input type="email" id="email" placeholder="Enter email address">

</div>

<div class="form-group">

<label for="mobile">Mobile Number</label>

<input type="tel" id="mobile" placeholder="Enter 10-digit mobile number">

</div>

<div class="form-group">

<label for="age">Age</label>



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```
<input type="number" id="age" placeholder="Enter age (16-90)">
```

```
<span id="ageError" class="error-message"></span>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="plan">Membership Plan</label>
```

```
<select id="plan">
```

```
<option value="">Select a plan</option>
```

```
<option value="monthly">Monthly Pass</option>
```

```
<option value="quarterly">Quarterly Pass (3 Months)</option>
```

```
<option value="annual">Annual Membership (1 Year)</option>
```

```
</select>
```

```
<span id="planError" class="error-message"></span>
```

```
</div>
```

```
<button type="submit">Register Now</button>
```

```
</form>
```

```
</div>
```

```
<script>
```

```
// 1. Access elements using document.getElementById
```

```
const nameInput = document.getElementById("name");
```



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```
const emailInput = document.getElementById("email");

const mobileInput = document.getElementById("mobile");

const ageInput = document.getElementById("age");

const planSelect = document.getElementById("plan");

const gymForm = document.getElementById("gymForm");


const nameError = document.getElementById("nameError");
const emailError = document.getElementById("emailError");
const mobileError = document.getElementById("mobileError");
const ageError = document.getElementById("ageError");
const planError = document.getElementById("planError");


// Helper functions for validation logic

function validateName() {

    // 2. Get input value

    const value = nameInput.value.trim();

    // 5. Validate condition & 4. Show message element

    if (value.length < 3) {

        nameError.textContent = "Name must be at least 3 characters";

        return false;

    }

    nameError.textContent = "";
```



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```
return true;

}

function validateEmail() {

    const value = emailInput.value.trim();

    const emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (!value.match(emailPattern)) {

        emailError.textContent = "Enter a valid email address";

        return false;

    }

    emailError.textContent = "";

    return true;

}

function validateMobile() {

    const value = mobileInput.value.trim();

    const mobilePattern = /^[0-9]{10}$/;

    if (!value.match(mobilePattern)) {

        mobileError.textContent = "Enter a valid 10-digit mobile number";

        return false;

    }

    mobileError.textContent = "";
```



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```
return true;

}

function validateAge() {

    const value = parseInt(ageInput.value, 10);

    if (isNaN(value) || value < 16 || value > 90) {

        ageError.textContent = "Age must be between 16 and 90";

        return false;

    }

    ageError.textContent = "";

    return true;

}

function validatePlan() {

    if (planSelect.value === "") {

        planError.textContent = "Please select a membership plan";

        return false;

    }

    planError.textContent = "";

    return true;

}
```



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```
// 3. Add event listeners for real-time validation ("input" & "change")

nameInput.addEventListener("input", validateName);

emailInput.addEventListener("input", validateEmail);

mobileInput.addEventListener("input", validateMobile);

ageInput.addEventListener("input", validateAge);

planSelect.addEventListener("change", validatePlan);


// 6. Stop form submission if validation fails

gymForm.addEventListener("submit", function(event) {

    const isNameValid = validateName();

    const isEmailValid = validateEmail();

    const isMobileValid = validateMobile();

    const isAgeValid = validateAge();

    const isPlanValid = validatePlan();


    if (!isNameValid || !isEmailValid || !isMobileValid || !isAgeValid || !isPlanValid) {

        // 6. Stop form submission

        event.preventDefault();

    } else {

        alert("Form submitted successfully!");

    }

});
```



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</script>

</body>

</html>

Output:

The screenshot shows a web browser window with the title 'Gym Membership Form'. The address bar displays the URL '127.0.0.1:5500/Experiments/24070521147/gym_admission_form.html'. A dark overlay message from '127.0.0.1:5500' states 'Form submitted successfully!' with an 'OK' button. The form itself contains the following fields:

- Name:** Tejas Santosh Bawane
- Email Address:** tejas.bawane.batch2024@sitnagpur.siu.edu.in
- Mobile Number:** 7498875903
- Age:** 20
- Membership Plan:** Monthly Pass (dropdown menu)
- Register Now:** A green button at the bottom of the form.



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Case Study No.: 08

Title: Online Job application form.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Form Handling with JavaScript</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
margin: 30px;
```

```
}
```

```
.form-container {
```

```
border: 1px solid #000;
```

```
width: 450px;
```

```
padding: 20px;
```

```
box-sizing: border-box;
```

```
}
```



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```
.mandatory-note {  
  color: red;  
  font-size: 13px;  
  margin-bottom: 15px;  
}
```

```
.form-row {  
  display: flex;  
  align-items: center;  
  margin-bottom: 10px;  
}
```

```
.form-row label {  
  width: 110px;  
  font-size: 14px;  
}
```

```
.form-row input[type="text"],  
.form-row input[type="email"],  
.form-row input[type="url"],  
.form-row input[type="password"],
```



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```
.form-row select {  
  padding: 3px;  
  border: 1px solid #777;  
}
```

```
.input-field {  
  width: 180px;  
}
```

```
.asterisk {  
  color: red;  
  margin-left: 4px;  
}
```

```
.date-selects select {  
  margin-right: 2px;  
}
```

```
.checkbox-row {  
  display: flex;  
  align-items: center;  
  margin-top: 15px;
```



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```
margin-bottom: 15px;

}
```

```
.checkbox-row label {

margin-left: 5px;

font-size: 14px;

}
```

```
.submit-btn {

padding: 4px 12px;

font-weight: bold;

cursor: pointer;

}
```

```
/* Input highlighting on focus event */
```

```
input:focus, select:focus {

background-color: #f0f8ff;

outline: 1px solid #0056b3;

}
```

```
</style>
```

```
</head>
```

```
<body>
```



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```
<div class="form-container">
```

```
<div class="mandatory-note">*Mandatory to fill</div>
```

```
<form id="signupForm" novalidate>
```

```
<div class="form-row">
```

```
<label for="firstname">Firstname:</label>
```

```
<input type="text" id="firstname" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="lastname">Lastname:</label>
```

```
<input type="text" id="lastname" class="input-field">
```

```
</div>
```

```
<div class="form-row">
```

```
<label>Birthday:</label>
```

```
<div class="date-selects">
```

```
<select id="day">
```

```
<option value="">Day</option>
```



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</select>

<select id="month">

<option value="">Month</option>

<option value="1">Jan</option>

<option value="2">Feb</option>

<option value="3">Mar</option>

<option value="4">Apr</option>

<option value="5">May</option>

<option value="6">Jun</option>

<option value="7">Jul</option>

<option value="8">Aug</option>

<option value="9">Sep</option>

<option value="10">Oct</option>

<option value="11">Nov</option>

<option value="12">Dec</option>

</select>

<select id="year">

<option value="">Year</option>

</select>

</div>

*

</div>



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```
<div class="form-row">
```

```
<label for="username">Username:</label>
```

```
<input type="text" id="username" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="email">E-mail:</label>
```

```
<input type="email" id="email" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="website">Website:</label>
```

```
<input type="url" id="website" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password" class="input-field">
```



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*

</div>

<div class="form-row">

<label for="repassword">Re-password:</label>

<input type="password" id="repassword" class="input-field">

*

</div>

<div class="checkbox-row">

<input type="checkbox" id="terms">

<label for="terms">I agree to the terms & conditions.</label>

</div>

<button type="submit" class="submit-btn">Sign up</button>

</form>

</div>

<script>

// Populate Days (1 to 31)

const daySelect = document.getElementById('day');

for (let i = 1; i <= 31; i++) {



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```
const opt = document.createElement('option');

opt.value = i;

opt.textContent = i;

daySelect.appendChild(opt);
}

// Populate Years (1950 to 2026)

const yearSelect = document.getElementById('year');

for (let i = 2026; i >= 1950; i--) {

    const opt = document.createElement('option');

    opt.value = i;

    opt.textContent = i;

    yearSelect.appendChild(opt);
}

// Event Listener: Change event demonstration

document.getElementById('month').addEventListener('change', function() {

    console.log('Month changed to: ' + this.value);

});

// Event Listener: Submit event and field validations

document.getElementById('signupForm').addEventListener('submit', function(e) {
```



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```
e.preventDefault();
```

```
const firstname = document.getElementById('firstname').value.trim();
```

```
const day = document.getElementById('day').value;
```

```
const month = document.getElementById('month').value;
```

```
const year = document.getElementById('year').value;
```

```
const username = document.getElementById('username').value.trim();
```

```
const email = document.getElementById('email').value.trim();
```

```
const website = document.getElementById('website').value.trim();
```

```
const password = document.getElementById('password').value;
```

```
const repassword = document.getElementById('repassword').value;
```

```
const terms = document.getElementById('terms').checked;
```

```
// Check empty fields
```

```
if (!firstname) {
```

```
    alert('Firstname is required.');
```

```
    return;
```

```
}
```

```
if (!day || !month || !year) {
```

```
    alert('Please select a complete birthday.');
```

```
    return;
```



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```
}
```

```
if (!username) {  
    alert('Username is required.');
```

```
    return;
```

```
}
```

```
// Email validation
```

```
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (!emailRegex.test(email)) {
```

```
    alert('Please enter a valid email address.');
```

```
    return;
```

```
}
```

```
// Website validation
```

```
const urlRegex = /^(https?:\/\/)?([w\d-]+\.)+[w-]+(\.*)?$/i;
```

```
if (!urlRegex.test(website)) {
```

```
    alert('Please enter a valid website URL.');
```

```
    return;
```

```
}
```

```
// Password validations
```



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```
if (!password) {  
    alert('Password is required.');
```



```
    return;  
}  
  
if (password !== repassword) {  
    alert('Passwords do not match.');
```



```
    return;  
}  
  
// Terms & Conditions checkbox check  
if (!terms) {  
    alert('You must agree to the terms & conditions.');
```



```
    return;  
}  
  
    alert('Form submitted successfully!');
```



```
});  
</script>  
  
</body>  
</html>
```



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Output:

Online Job Application Form

127.0.0.1:5500/Case%20Study/24070521212/job_application.html

Online Job Application Form

Scenario: Please fill in all fields correctly before submission.

Personal Information

Full Name
Aishwarya Brahmarkar

Date of Birth
06/02/2006

Gender
Female

Email Address
aishwarya.brahmarkar.batch2024@sitnagpur.siu.edu.in

Mobile Number
9617305200

Address
C-21, Kristu Nagar, near St. Joseph school, Amlidih, Raipur, Chhattisgarh

City / State / PIN Code
Raipur

Educational Qualification



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Experiment No.: 09

Title: Implement localStorage and sessionStorage to save user preferences like theme settings.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Theme Settings</title>

  <style>

    /* Default Page Layout */

    body {

      font-family: Arial, sans-serif;

      display: flex;

      justify-content: center;

      align-items: center;

      min-height: 100vh;

      margin: 0;

      background-color: #f4f4f9;

      transition: background-color 0.3s, color 0.3s;

    }
```



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/* Container Card */

```
.theme-card {  
  border: 2px solid #555;  
  padding: 30px;  
  width: 320px;  
  text-align: center;  
  border-radius: 4px;  
  transition: background-color 0.3s, border-color 0.3s;  
}
```

/* Light Theme Styling */

```
body.light-theme {  
  background-color: #f4f4f9;  
  color: #000000;  
}  
  
body.light-theme .theme-card {  
  background-color: #ffffff;  
  border-color: #888;  
}
```

/* Dark Theme Styling */



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```
body.dark-theme {  
  
  background-color: #121212;  
  
  color: #ffffff;  
  
}  
  
body.dark-theme .theme-card {  
  
  background-color: #38322e;  
  
  border-color: #666;  
  
}  
  
/* Form Controls & Buttons */  
  
select {  
  
  margin-top: 10px;  
  
  margin-bottom: 20px;  
  
  padding: 4px 8px;  
  
}  
  
.button-group {  
  
  display: flex;  
  
  justify-content: space-between;  
  
  gap: 10px;  
  
  margin-bottom: 15px;  
  
}
```



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```
button {  
  
  padding: 8px 12px;  
  
  cursor: pointer;  
  
  font-size: 13px;  
  
  border: 1px solid #777;  
  
  border-radius: 3px;  
  
  background-color: #efefef;  
  
  color: #000;  
  
}
```

```
button:hover {  
  
  background-color: #e0e0e0;  
  
}
```

```
.clear-btn {  
  
  margin-bottom: 20px;  
  
}
```

```
.status-msg {  
  
  font-size: 14px;  
  
  font-weight: bold;
```



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```
min-height: 20px;

}

</style>

</head>

<body class="light-theme">

  <div class="theme-card">

    <h2>Theme Settings</h2>

    <label for="themeSelect">Select your preferred theme:</label><br>

    <select id="themeSelect">

      <option value="light-theme">Light</option>

      <option value="dark-theme">Dark</option>

    </select>

    <div class="button-group">

      <button onclick="saveLocalStorage()">Save using localStorage</button>

      <button onclick="saveSessionStorage()">Save using sessionStorage</button>

    </div>

    <button class="clear-btn" onclick="clearPreferences()">Clear Preferences</button>
```



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```
<div class="status-msg" id="statusMessage"></div>
```

```
</div>
```

```
<script>
```

```
const themeSelect = document.getElementById('themeSelect');
```

```
const statusMessage = document.getElementById('statusMessage');
```

```
// 1. Change theme immediately when dropdown option is selected
```

```
themeSelect.addEventListener('change', (e) => {
```

```
    document.body.className = e.target.value;
```

```
});
```

```
// 2. Save preference to localStorage
```

```
function saveLocalStorage() {
```

```
    const selectedTheme = themeSelect.value;
```

```
    localStorage.setItem('preferredTheme', selectedTheme);
```

```
    // Remove any sessionStorage theme to avoid conflicts
```

```
    sessionStorage.removeItem('preferredTheme');
```

```
    document.body.className = selectedTheme;
```

```
    statusMessage.innerText = 'Theme saved using localStorage.';
```

```
}
```



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// 3. Save preference to sessionStorage

```
function saveSessionStorage() {  
    const selectedTheme = themeSelect.value;  
    sessionStorage.setItem('preferredTheme', selectedTheme);  
    // Remove any localStorage theme to avoid conflicts  
    localStorage.removeItem('preferredTheme');  
  
    document.body.className = selectedTheme;  
    statusMessage.innerText = 'Theme saved using sessionStorage.';  
}
```

// 4. Clear all saved preferences

```
function clearPreferences() {  
    localStorage.removeItem('preferredTheme');  
    sessionStorage.removeItem('preferredTheme');  
  
    // Reset back to default (Light)  
    document.body.className = 'light-theme';  
    themeSelect.value = 'light-theme';  
    statusMessage.innerText = 'Preferences cleared.';  
}
```



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```
// 5. Load saved theme when the page loads

window.addEventListener('DOMContentLoaded', () => {

    const localTheme = localStorage.getItem('preferredTheme');

    const sessionTheme = sessionStorage.getItem('preferredTheme');

    if (localTheme) {

        document.body.className = localTheme;

        themeSelect.value = localTheme;

        statusMessage.innerText = 'Theme loaded from localStorage.';

    } else if (sessionTheme) {

        document.body.className = sessionTheme;

        themeSelect.value = sessionTheme;

        statusMessage.innerText = 'Theme loaded from sessionStorage.';

    }

});

</script>

</body>

</html>
```



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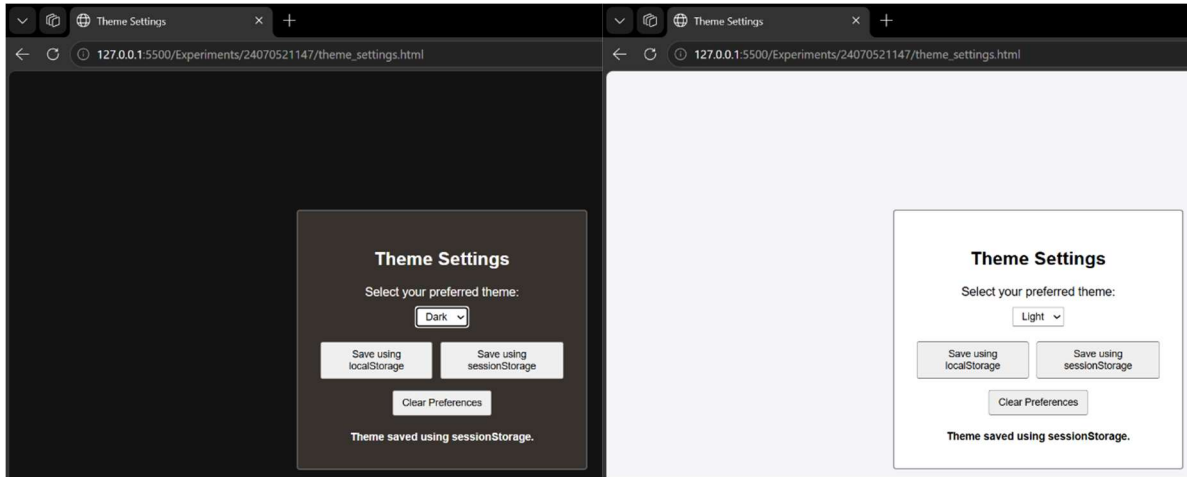
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Output:





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Case Study No.: 09

Title: Apply theme toggle logic to all experiments 1-8.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Form Handling with JavaScript</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      margin: 30px;

    }

    .form-container {

      border: 1px solid #000;

      width: 450px;

      padding: 20px;

      box-sizing: border-box;

    }
```



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```
.mandatory-note {  
  color: red;  
  font-size: 13px;  
  margin-bottom: 15px;  
}
```

```
.form-row {  
  display: flex;  
  align-items: center;  
  margin-bottom: 10px;  
}
```

```
.form-row label {  
  width: 110px;  
  font-size: 14px;  
}
```

```
.form-row input[type="text"],  
.form-row input[type="email"],  
.form-row input[type="url"],  
.form-row input[type="password"],
```



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```
.form-row select {  
  padding: 3px;  
  border: 1px solid #777;  
}
```

```
.input-field {  
  width: 180px;  
}
```

```
.asterisk {  
  color: red;  
  margin-left: 4px;  
}
```

```
.date-selects select {  
  margin-right: 2px;  
}
```

```
.checkbox-row {  
  display: flex;  
  align-items: center;  
  margin-top: 15px;
```



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```
margin-bottom: 15px;

}
```

```
.checkbox-row label {

margin-left: 5px;

font-size: 14px;

}
```

```
.submit-btn {

padding: 4px 12px;

font-weight: bold;

cursor: pointer;

}
```

```
/* Input highlighting on focus event */
```

```
input:focus, select:focus {

background-color: #f0f8ff;

outline: 1px solid #0056b3;

}
```

```
</style>
```

```
</head>
```

```
<body>
```



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```
<div class="form-container">
```

```
<div class="mandatory-note">*Mandatory to fill</div>
```

```
<form id="signupForm" novalidate>
```

```
<div class="form-row">
```

```
<label for="firstname">Firstname:</label>
```

```
<input type="text" id="firstname" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="lastname">Lastname:</label>
```

```
<input type="text" id="lastname" class="input-field">
```

```
</div>
```

```
<div class="form-row">
```

```
<label>Birthday:</label>
```

```
<div class="date-selects">
```

```
<select id="day">
```

```
<option value="">Day</option>
```



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</select>

<select id="month">

<option value="">Month</option>

<option value="1">Jan</option>

<option value="2">Feb</option>

<option value="3">Mar</option>

<option value="4">Apr</option>

<option value="5">May</option>

<option value="6">Jun</option>

<option value="7">Jul</option>

<option value="8">Aug</option>

<option value="9">Sep</option>

<option value="10">Oct</option>

<option value="11">Nov</option>

<option value="12">Dec</option>

</select>

<select id="year">

<option value="">Year</option>

</select>

</div>

*

</div>



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```
<div class="form-row">
```

```
<label for="username">Username:</label>
```

```
<input type="text" id="username" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="email">E-mail:</label>
```

```
<input type="email" id="email" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="website">Website:</label>
```

```
<input type="url" id="website" class="input-field">
```

```
<span class="asterisk">*</span>
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="password">Password:</label>
```

```
<input type="password" id="password" class="input-field">
```



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*

</div>

<div class="form-row">

<label for="repassword">Re-password:</label>

<input type="password" id="repassword" class="input-field">

*

</div>

<div class="checkbox-row">

<input type="checkbox" id="terms">

<label for="terms">I agree to the terms & conditions.</label>

</div>

<button type="submit" class="submit-btn">Sign up</button>

</form>

</div>

<script>

// Populate Days (1 to 31)

const daySelect = document.getElementById('day');

for (let i = 1; i <= 31; i++) {



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```
const opt = document.createElement('option');

opt.value = i;

opt.textContent = i;

daySelect.appendChild(opt);
}

// Populate Years (1950 to 2026)

const yearSelect = document.getElementById('year');

for (let i = 2026; i >= 1950; i--) {

    const opt = document.createElement('option');

    opt.value = i;

    opt.textContent = i;

    yearSelect.appendChild(opt);
}

// Event Listener: Change event demonstration

document.getElementById('month').addEventListener('change', function() {

    console.log('Month changed to: ' + this.value);

});

// Event Listener: Submit event and field validations

document.getElementById('signupForm').addEventListener('submit', function(e) {
```



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```
e.preventDefault();
```

```
const firstname = document.getElementById('firstname').value.trim();
```

```
const day = document.getElementById('day').value;
```

```
const month = document.getElementById('month').value;
```

```
const year = document.getElementById('year').value;
```

```
const username = document.getElementById('username').value.trim();
```

```
const email = document.getElementById('email').value.trim();
```

```
const website = document.getElementById('website').value.trim();
```

```
const password = document.getElementById('password').value;
```

```
const repassword = document.getElementById('repassword').value;
```

```
const terms = document.getElementById('terms').checked;
```

```
// Check empty fields
```

```
if (!firstname) {
```

```
    alert('Firstname is required.');
```

```
    return;
```

```
}
```

```
if (!day || !month || !year) {
```

```
    alert('Please select a complete birthday.');
```

```
    return;
```



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```
}
```

```
if (!username) {  
    alert('Username is required.');
```

```
    return;
```

```
}
```

```
// Email validation
```

```
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (!emailRegex.test(email)) {
```

```
    alert('Please enter a valid email address.');
```

```
    return;
```

```
}
```

```
// Website validation
```

```
const urlRegex = /^(https?:\/\/)?([\w\d-]+\.[\w-]+(\w.*)?)$/i;
```

```
if (!urlRegex.test(website)) {
```

```
    alert('Please enter a valid website URL.');
```

```
    return;
```

```
}
```

```
// Password validations
```



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```
if (!password) {  
    alert('Password is required.');
```



```
    return;  
}  
  
if (password !== repassword) {  
    alert('Passwords do not match.');
```



```
    return;  
}  
  
// Terms & Conditions checkbox check  
if (!terms) {  
    alert('You must agree to the terms & conditions.');
```



```
    return;  
}  
  
    alert('Form submitted successfully!');  
});  
</script>  
  
</body>  
</html>
```



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Output:

The image displays two side-by-side screenshots of a web browser showing a 'Gym Membership Form'. The left screenshot shows the form in a dark theme, and the right screenshot shows the form in a light theme. Both forms are titled 'Gym Membership Form' and contain the following fields:

- Full Name: Aishwarya Brahmanekar
- Email Address: aishwarya.brahmanekar.batch2024@sitnagpur.siu.edu.in
- Mobile Number: 9617305200
- Age: 20
- Membership Plan: Monthly Pass (selected from a dropdown menu)
- Register Now (button)

The browser address bar shows the URL: 127.0.0.1:5500/Case%20Study/24070521212/Gym_membership_form.html.



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Experiment No.: 10

Title: Load and display JSON data using fetch() and jQuery \$.getJSON(); render results dynamically in a table.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Indian Student Directory</title>

  <!-- jQuery CDN included for $.getJSON functionality -->

  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

  <style>

    body { font-family: sans-serif; padding: 20px; }

    table { width: 100%; border-collapse: collapse; margin-top: 15px; }

    th, td { border: 1px solid #ccc; padding: 8px 12px; text-align: left; }

    th { background-color: #f4f4f4; }

    button { padding: 8px 16px; margin-right: 8px; cursor: pointer; }

  </style>

</head>

<body>
```



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<h2>Indian Student Directory</h2>

<button onclick="loadWithFetch()">Load using fetch()</button>

<button onclick="loadWithJQuery()">Load using \$.getJSON()</button>

<table>

<thead>

<tr>

<th>Student ID</th>

<th>Name</th>

<th>Email</th>

</tr>

</thead>

<tbody id="table-body">

<!-- Dynamic rows load here -->

</tbody>

</table>

<script>

// Indian student data with IDs starting from 101

const studentData = [

{ id: 101, name: "Aarav Sharma", email: "aarav.sharma@example.in" },

{ id: 102, name: "Ananya Patel", email: "ananya.patel@example.in" },



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```
{ id: 103, name: "Rohan Verma", email: "rohan.verma@example.in" },
{ id: 104, name: "Priya Iyer", email: "priya.iyer@example.in" },
{ id: 105, name: "Aditya Kulkarni", email: "aditya.kulkarni@example.in" },
{ id: 106, name: "Diya Nair", email: "diya.nair@example.in" }
];

// Helper function to build and inject rows
function renderRows(data) {
  const tbody = document.getElementById('table-body');
  tbody.innerHTML = ""; // Clear table

  data.forEach(student => {
    const row = `<tr>
      <td>${student.id}</td>
      <td>${student.name}</td>
      <td>${student.email}</td>
    </tr>`;
    tbody.innerHTML += row;
  });
}

// 1. fetch() simulation using a Promise
```



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```
function loadWithFetch() {  
  
    // Simulating a fetch API request with a resolved Promise  
  
    Promise.resolve(studentData)  
  
        .then(data => renderRows(data))  
  
        .catch(err => console.error('Fetch error:', err));  
  
}
```

// 2. jQuery simulation using Deferred object

```
function loadWithjQuery() {  
  
    // Simulating $.getJSON with a jQuery Deferred object  
  
    $.Deferred().resolve(studentData)  
  
        .done(function(data) {  
  
            renderRows(data);  
  
        })  
  
        .fail(function(err) {  
  
            console.error('jQuery error:', err);  
  
        });  
  
}  
  
</script>
```

</body>

</html>



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Output:

Indian Student Directory

127.0.0.1:5500/Experiments/24070521212/json-table.html

Indian Student Directory

Load using fetch()

Load using \$.getJSON()

| Student ID | Name | Email |
|------------|-----------------|----------------------------|
| 101 | Aarav Sharma | aarav.sharma@example.in |
| 102 | Ananya Patel | ananya.patel@example.in |
| 103 | Rohan Verma | rohan.verma@example.in |
| 104 | Priya Iyer | priya.iyer@example.in |
| 105 | Aditya Kulkarni | aditya.kulkarni@example.in |
| 106 | Diya Nair | diya.nair@example.in |



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Case Study No.: 10

Title: Campus Event Registration System.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

1. Index.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Campus Event Registration System</title>

  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

  <style>

    body { font-family: Arial, sans-serif; margin: 20px; }

    .controls { margin-bottom: 15px; display: flex; gap: 10px; }

    input, select, button { padding: 8px; }

    table { width: 100%; border-collapse: collapse; margin-top: 15px; }

    th, td { border: 1px solid #ccc; padding: 10px; text-align: left; }

    th { background-color: #f4f4f4; }

    .message { color: red; margin-top: 15px; font-weight: bold; }

  </style>

</head>

<body>
```



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<h2>Campus Event Registration System</h2>

<div class="controls">

<button id="loadBtn">Load Student Data</button>

<input type="text" id="searchInput" placeholder="Search by Name or PRN...">

<select id="statusFilter">

<option value="All">All Statuses</option>

<option value="Registered">Registered</option>

<option value="Pending">Pending</option>

<option value="Cancelled">Cancelled</option>

</select>

</div>

<div id="message" class="message"></div>

<table id="studentTable">

<thead>

<tr>

<th>Student Name</th>

<th>PRN</th>

<th>Department</th>



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```
<th>Year</th>

<th>Event Name</th>

<th>Registration Status</th>

</tr>

</thead>

<tbody id="tableBody"></tbody>

</table>

<script src="script.js"></script>

</body>

</html>
```

2. Script.js

```
3. let studentsData = [];
4.
5. // Fetch data using JavaScript fetch API
6. $('#loadBtn').on('click', function() {
7.   fetch('students.json')
8.     .then(response => {
9.       if (!response.ok) throw new Error('Network response was not ok');
10.      return response.json();
11.    })
12.    .then(data => {
13.      studentsData = data;
14.      $('#message').text('');
15.      renderTable(studentsData);
16.    })
17.    .catch(() => {
18.      $('#tableBody').empty();
19.      $('#message').text('JSON data cannot be loaded.');
```



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```
22.
23. /*
24. // Alternative implementation using jQuery $.getJSON()
25. $('#loadBtn').on('click', function() {
26.     $.getJSON('students.json')
27.     .done(function(data) {
28.         studentsData = data;
29.         $('#message').text('');
30.         renderTable(studentsData);
31.     })
32.     .fail(function() {
33.         $('#tableBody').empty();
34.         $('#message').text('JSON data cannot be loaded.');
```



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```
60. function renderTable(data) {
61.   const $tbody = $('#tbody');
62.   $tbody.empty();
63.
64.   data.forEach(student => {
65.     const row = `<tr>
66.       <td>${student.studentName}</td>
67.       <td>${student.prn}</td>
68.       <td>${student.department}</td>
69.       <td>${student.year}</td>
70.       <td>${student.eventName}</td>
71.       <td>${student.registrationStatus}</td>
72.     </tr>`;
73.     $tbody.append(row);
74.   });
75. }
76.
77. // Event Listeners for Live Search & Filter
78. $('#searchInput').on('input', filterAndRender);
79. $('#statusFilter').on('change', filterAndRender);
```

3. students.json

```
[
  {
    "studentName": "AISHWARYA BRAHMANKAR",
    "prn": "24070521212",
    "department": "Computer Science",
    "year": "3rd",
    "eventName": "Hackathon 2026",
    "registrationStatus": "Registered"
  },
]
```



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```
{  
  
  "studentName": "ASWIN BINU",  
  
  "prn": "24070521222",  
  
  "department": "Information Technology",  
  
  "year": "2nd",  
  
  "eventName": "AI Summit",  
  
  "registrationStatus": "Pending"  
},  
  
{  
  
  "studentName": "BHAISARE OJAS JAGDISH",  
  
  "prn": "24070521224",  
  
  "department": "Mechanical",  
  
  "year": "4th",  
  
  "eventName": "RoboWars",  
  
  "registrationStatus": "Cancelled"  
},  
  
{  
  
  "studentName": "BHOYAR RITIKA PRAKASH",  
  
  "prn": "24070521225",  
  
  "department": "Computer Science",  
  
  "year": "3rd",  
  
  "eventName": "Hackathon 2026",
```



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```
"registrationStatus": "Registered"

},

{

  "studentName": "BHUMIKA ARORA",

  "prn": "24070521227",

  "department": "Information Technology",

  "year": "2nd",

  "eventName": "AI Summit",

  "registrationStatus": "Pending"

},

{

  "studentName": "BUTLE SHARWARI ABHAY",

  "prn": "24070521228",

  "department": "Mechanical",

  "year": "4th",

  "eventName": "RoboWars",

  "registrationStatus": "Cancelled"

},

{

  "studentName": "DAWARE PARTH CHANDRASHEKAR",

  "prn": "24070521230",

  "department": "Computer Science",
```



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```
"year": "3rd",

"eventName": "Hackathon 2026",

"registrationStatus": "Registered"

},

{

"studentName": "GORE ADITI VAIBHAV",

"prn": "24070521233",

"department": "Information Technology",

"year": "2nd",

"eventName": "AI Summit",

"registrationStatus": "Pending"

},

{

"studentName": "JASH CHAUHAN",

"prn": "24070521242",

"department": "Mechanical",

"year": "4th",

"eventName": "RoboWars",

"registrationStatus": "Cancelled"

}

]
```



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Output:

Campus Event Registration System

127.0.0.1:5500/Case%20Study/24070521212/index.html

Campus Event Registration System

Load Student Data

Search by Name or PRN...

All Statuses

Student Name	PRN	Department	Year	Event Name	Registration Status
AISHWARYA BRAHMANKAR	24070521212	Computer Science	3rd	Hackathon 2026	Registered
ASWIN BINU	24070521222	Information Technology	2nd	AI Summit	Pending
BHAISARE OJAS JAGDISH	24070521224	Mechanical	4th	RoboWars	Cancelled
BHOYAR RITIKA PRAKASH	24070521225	Computer Science	3rd	Hackathon 2026	Registered
BHUMIKA ARORA	24070521227	Information Technology	2nd	AI Summit	Pending
BUTLE SHARWARI ABHAY	24070521228	Mechanical	4th	RoboWars	Cancelled
DAWARE PARTH CHANDRASHEKAR	24070521230	Computer Science	3rd	Hackathon 2026	Registered
GORE ADITI VAIBHAV	24070521233	Information Technology	2nd	AI Summit	Pending
JASH CHAUHAN	24070521242	Mechanical	4th	RoboWars	Cancelled

Campus Event Registration System

127.0.0.1:5500/Case%20Study/24070521212/index.html

Campus Event Registration System

Load Student Data

24070521212

All Statuses

Student Name	PRN	Department	Year	Event Name	Registration Status
AISHWARYA BRAHMANKAR	24070521212	Computer Science	3rd	Hackathon 2026	Registered